Information Technologies for Industrial Engineers

JavaScript Frameworks

What is a framework?

- A framework is a library that offers **opinions** about how software gets built.
- These opinions allow for **predictability** and **homogeneity** in an application.
  - Predictability allows software to scale to an enormous size and still be maintainable.
  - Predictability and maintainability are essential for the health and longevity of software.

Imperative and declarative programming

- **Imperative**
  - Tell a computer how to do something.
- **Declarative**
  - Describe to a computer what you want.

To do this

- List 1
- List 2
- List 3

Imperative programming

- JavaScript

```javascript
const ul = document.createElement("ul");
for (let i = 1; i < 4; i++) {
  const li = document.createElement("li");
  li.textContent = `List ${i}`;
  ul.appendChild(li);
}
document.body.appendChild(ul);
```

Declarative

- Just use HTML

```html
<ul>
  <li>List 1</li>
  <li>List 2</li>
  <li>List 3</li>
</ul>
```

But...

- `HTML` only works for static site.
- For dynamic web application, a **framework** can help us write code more **declaratively**.

Vanilla JS or Framework?

- Vanilla JS
  - A less complex site, for example a personal to-do list or a site that displays mostly static content
- Framework
  - A large site with a complex UI. Frameworks provide solutions to common problems that would take an absurd amount of time and patience to implement with pure JavaScript.

Framework popularity

- State of JS 2025
- Stack Overflow Survey 2025

Vue

Vue Template Syntax

```
<template>
  <h1>Hello, world!</h1>
</template>
```

- Vue uses template syntax to describe UI declaratively.

Vue Single-File Component (SFC)

- A Vue SFC is a file that encapsulates the template, logic, and styles of a component in a single file.

```
<script setup lang="ts">
import { ref } from "vue";

const count = ref(0);
</script>

<template>
  <button @click="count++">Count: {{ count }}</button>
</template>
```

Comparison with Vanilla JS

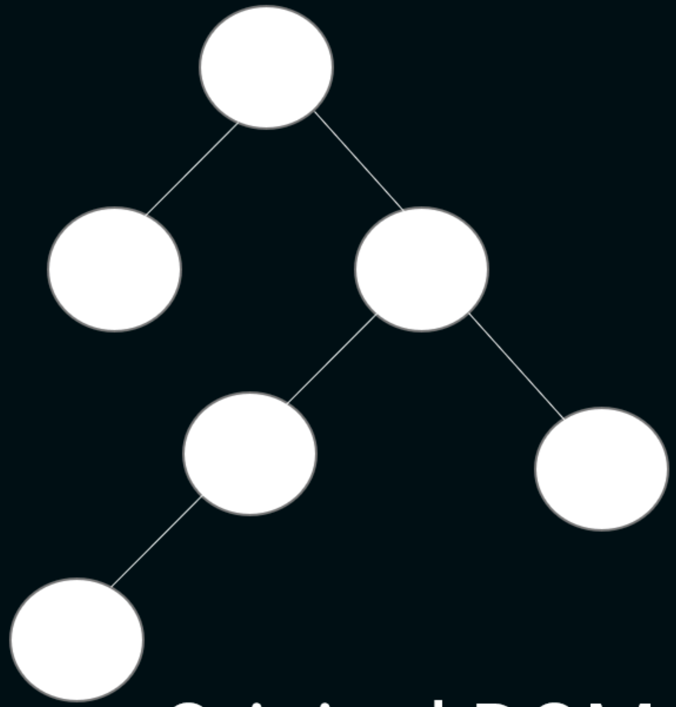
```html
<button>Count: 0</button>
```

```javascript
const btn = document.querySelector("button");
let count = 0;
function updateButton() {
  btn.textContent = `Count: ${count}`;
}
btn.addEventListener("click", () => {
  count++;
  updateButton();
});
```

Link to CodePen

Virtual DOM

- Virtual DOM (VDOM) represents the DOM tree in memory.
- VDOM performs calculations and decides which elements of the actual DOM should be mutated.
    - Faster performance.
- Source



Original DOM

Rendering process

- Vue uses a **reactive system** to track changes in the state of your application.
- When a reactive state changes, Vue automatically re-renders the affected components.
- Vue tracks dependencies and updates only the affected parts of the DOM.
- See illustration in Source.

Get started with `vue`

- `pnpm create vite@latest`
  - `Project name`: my-app (any name)
  - `Select a framework`: Vue
  - `Select a variant`: TypeScript
- `cd my-app`
- `code . -r` *(Set VSCode workspace)*
- `pnpm install`
- `pnpm run dev`

Build

- `pnpm run build`

Preview

- `pnpm run preview`

Deploy

- Deploy `./dist` folder to Netlify.

Online IDE

- https://codesandbox.io/
- https://stackblitz.com/